

Homework for week 1, submission deadline 23:59 on Sunday night.

1. Complete an Omics-to-Question map: Using your own research project or biological interest. What kind of omics you plan to use to solve the biological research question?

研究问题：藻类在高光胁迫环境下，通过哪些分子机制实现光保护响应，抵御强光带来的氧化损伤？

选用组学：

①转录组学：解析强光处理前后藻类基因表达水平的变化，找到显著上调 / 下调的基因，挖掘参与光保护、抗氧化通路的候选基因，回答哪些基因在转录层面参与高光胁迫应答。

②蛋白质组学：检测胁迫条件下蛋白质丰度与蛋白修饰的改变。基因转录不等于蛋白表达，蛋白组可以反映实际执行生理功能的蛋白，回答哪些功能蛋白参与光保护反应。

③代谢组学：测定胞内代谢物如抗氧化物质、光合中间产物等含量变化，反映细胞生理表型，回答高光胁迫下细胞代谢产物如何重编程，以此对抗氧化损伤。

2. Write one weak prompt and one CARE+V prompt: For the same biological or research question, propose a weak and strong (CARE+V) prompt that can be tested by AI. Use any LLMs to test your prompt and paste the output results below.

Weak prompt：讲讲藻类的光抑制现象

Output：

讲讲藻类的光抑制现象

已思考 (用时 3 秒)

用户想了解藻类的光抑制现象。我需要提供一份全面的解释，涵盖定义、机制、影响因素等。为了全面获取信息，我将同时进行多项搜索，以覆盖不同角度和潜在的相关内容。

搜索到 45 个网页

搜索结果显示了一些相关页面。为了全面获取信息，我需要同时打开这些页面。

浏览 6 个页面

Photoinhibition of photosynthesis in Macrocyctis pyrifera (Phaeophyceae), Chondrus crispus (Rhodophyceae) an...
Light and oxygen stress in <i>Spirulina platensis</i> (cyanobacteria) grown outdoors in tubular reactors - Light a...
光合作用是藻类的基础。最重要的生理活动。能够快速对紫外辐射(Ultraviolet radiation, UVR)作出响应。
发现抑制光的光合生物对极端光照的抗性机制。
Photoinhibition of Photosynthesis and Its Recovery in Low and High Light Acclimatized Blue-green Algae (Cyano...
查看全部

这些资料提供了关于光抑制的详细信息。我的回答将涵盖定义、机制、影响因素、藻类的适应与修复机制、以及研究意义等方面。我会引用多个来源来支持我的回答。现在开始组织回答。

藻类的光抑制现象，简单来说，就是当藻类吸收的光能超过其光合作用所能利用的限度时，光合效率不升反降，甚至发生损伤的现象。

可以把这理解成一个“能量过载”的过程。虽然光是光合作用的驱动力，但“过犹不及”，过量的光能对藻类的光合系统造成压力。

藻类的防御与修复：损伤与修复的动态平衡

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藻类并非被动承受伤害，它们进化出了一套复杂的防御和修复系统。光抑制的最终程度，取决于损伤速率与修复速率之间的平衡。主要机制包括：

- 动态光抑制：这是一种可逆的、保护性的快速响应机制。藻类会通过非光化学淬灭 (NPQ)，将多余的光能以热量的形式安全地耗散掉。
- 修复循环：当PSII的D1蛋白受损后，藻类会启动一个精密的“修复循环”，迅速降解并重新合成全新的D1蛋白，以替换掉受损的部分，恢复PSII的功能。
- 长期适应：在长期高光环境下，藻类会发生适应性变化。例如，减小捕光天线尺寸以减少光能吸收，或积累保护性类胡萝卜素来淬灭ROS。

加剧光抑制的“帮凶”：环境因素

除了光照强度本身，一些环境因素会显著加剧光抑制：

- 紫外辐射 (UVR)：紫外辐射对藻类的抑制作用比可见光更严重，能直接损伤DNA等分子，并大幅增加ROS的产生。
- 低温：低温会减缓新陈代谢，包括D1蛋白的合成速率。因此，在低温强光下，修复速率跟不上损伤速率，光抑制现象会极其显著。
- 高氧浓度：高浓度的氧气会增加生成ROS的风险，与强光协同作用，加剧光抑制和光氧化伤害。
- 高温：极端高温本身会对细胞造成热损伤，与强光抑制叠加，对藻类造成双重打击。

研究光抑制的现实意义

理解光抑制现象具有重要的应用价值：

- 优化生物技术生产：在微藻大规模培养中，通过控制光强、光质或使用遮光材料来减轻光抑制，可显著提高生物产量。
- 理解生态变化：光抑制是影响藻类在水体中垂直分布和群落组成的关键因素。例如，适应弱光的深水物种被带到水面强光下时，会遭受严重的光抑制。

总而言之，藻类的光抑制现象是一个损伤与修复的动态博弈过程。理解这一机制，不仅有助于我们认识

CARE+V:

【C-Context 背景】微藻进行产氧光合作用。过高光照会引发光抑制，降低光合效率，损伤光系统 II (PSII)。回答面向本科分子生物学水平。

【A-Assignment 任务】解释藻类在强光下发生光抑制的分子机制。

【R-Requirements 要求】

- 区分光损伤过程和光保护过程。
- 必须提到 D1 蛋白的损伤与修复循环。
- 输出分为 3 个短段落。
- 专业名词附带简要解释，不要堆砌晦涩术语。

【E-Example 示例】第一段：简要定义光抑制；第二段：阐述光系统 II 中 D1 蛋白损伤机制；第三段：总结藻类主要光保护策略。

【V-Validation 校验】确认回答对象是藻类，不要大量讲陆生植物；确认已经覆盖 D1 蛋白周转这一核心知识点。

Output:

第一段：光抑制的定义与核心矛盾

光抑制是指藻类在接收的光能超过光合作用利用能力时，光合效率持续下降的现象，其本质是光系统 II (PSII，位于类囊体膜上的光合反应中心复合体) 的活性受损。强光下，PSII 的反应中心吸收过多激发能，导致电子传递链“堵塞”，产生高反应性的活性氧 (ROS，如单线态氧和过氧化氢)。这些 ROS 直接攻击 PSII 核心亚基，尤其破坏电子供体侧的氨基酸残基，使电荷分离和水的氧化反应受阻。这里需要区分两个并行但方向相反的过程：**光损伤** (物理化学性的不可逆破坏) 和 **光保护** (细胞主动启动的防御与修复机制)，两者的平衡决定了藻类能否在强光下存活。

第二段：D1 蛋白的损伤与修复循环——光抑制的核心分子事件

PSII 反应中心的核心催化亚基是 **D1 蛋白** (由叶绿体 *psbA* 基因编码，直接结合电子传递所需的醌分子和锰簇)。强光下，ROS 特异性地氧化 D1 蛋白上的组氨酸和甲硫氨酸残基，引发构象变化，导致 PSII 的电子供体侧与受体侧解耦联，光合电子传递中断。藻类应对此损伤的策略是“修复循环”：受损的 PSII 单体从类囊体膜的非堆叠区转移至堆叠区，随后 **FtsH 蛋白酶** (一种膜结合 ATP 依赖性金属蛋白酶) 识别并降解受损的 D1 蛋白，同时新的 D1 蛋白从叶绿体核糖体合成并插入空位，与未损伤的 CP43、CP47 等亚基重新组装成活性 PSII。这一循环在数分钟内即可启动，但若损伤速率超过新 D1 蛋白的合成速率，则修复滞后，光抑制加剧。藻类 (如莱茵衣藻) 中，D1 蛋白的周转还受叶绿体基因表达的可变剪接和翻译延伸因子调控，这与陆生植物的核基因调控有明显差异。

第三段：藻类特有的主要光保护策略

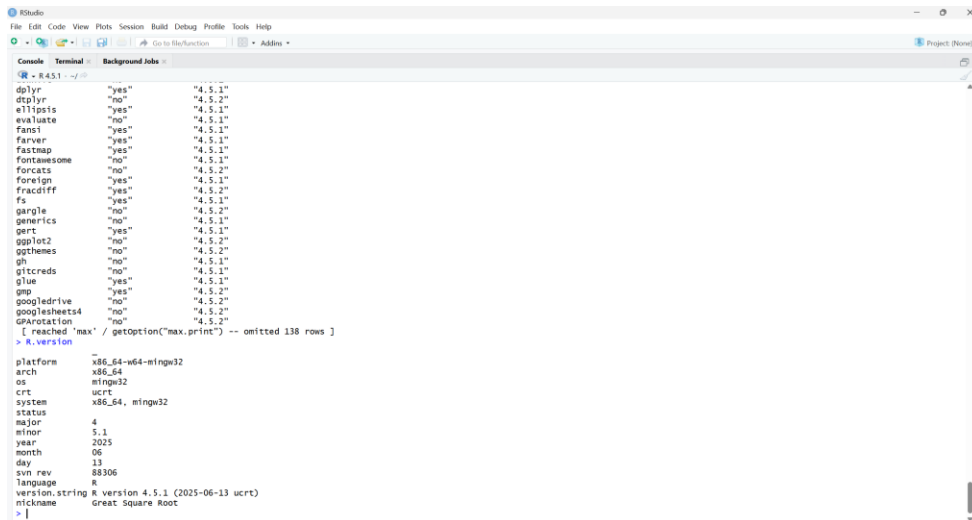
除了 D1 修复，藻类进化出多种主动光保护机制以预防损伤发生：① **非光化学淬灭 (NPQ)** ——通过类囊体膜上的 LHCII 捕光天线复合体构象改变，将过剩激发能以热能形式耗散，藻类的 NPQ 响应比陆生植物更快，且依赖于叶黄素循环 (如硅藻中的硅甲藻黄素)；② **状态转换 (state transition)** ——通过 LHCII 的可逆磷酸化，使部分捕光天线从 PSII 移动至光系统 I (PSI)，从而平衡两个光系统的激发能分配，减少 PSII 的过载；③ **抗氧化酶系统** ——超氧化物歧化酶 (SOD) 和抗坏血酸过氧化物酶 (APX) 迅速清除 ROS，降低 D1 蛋白的氧化攻击频率。值得注意的是，藻类在长期强光下还会下调 *psbA* 基因的翻译速率，主动降低新 D1 蛋白合成，转而积累热耗散蛋白 (如 L1818 家族)，这是一种“牺牲效率换取稳定”的生存策略，与陆生植物的避荫反应截然不同。综上，藻类的光抑制不是单纯的损坏，而是损伤—修复—预防三者动态博弈的结果。

3. Install and Run R and RStudio and save output showing R version and packages.

R: <https://cran.r-project.org/> RStudio: <https://posit.co/products/open-source/rstudio>.

Once installation, please paste a screen cut of Rstudio here. Give a try of R to using the foler (Week1_R_RStudio_Practice_Files) code and data.

Rstudio screen cut:



The screenshot shows the RStudio interface with the console pane active. It displays the output of several R commands. The first part shows the installation status of various packages, with columns for package name, whether it was installed, and the R version. The second part shows the output of the `R.version` command, which provides detailed information about the R environment, including platform, architecture, OS, CRT, system, status, major/minor version, year, month, day, SVN revision, language, version string, and nickname.

```
R - R 4.5.1 ~/>
dplyr      "yes"      "4.5.1"
displayr   "no"       "4.5.2"
ellipsis   "yes"      "4.5.1"
evaluate    "no"       "4.5.1"
Fansi       "yes"      "4.5.1"
farver      "yes"      "4.5.1"
fastmap     "yes"      "4.5.1"
fontawesome "no"       "4.5.1"
forcats     "no"       "4.5.2"
foreign     "yes"      "4.5.1"
fracdiff    "yes"      "4.5.2"
fs          "yes"      "4.5.1"
gargle      "no"       "4.5.2"
generics    "no"       "4.5.1"
gert        "yes"      "4.5.1"
ggplot2     "no"       "4.5.2"
ggthemes    "no"       "4.5.2"
gh          "no"       "4.5.1"
gitcreds    "no"       "4.5.1"
glue        "yes"      "4.5.1"
gmp         "yes"      "4.5.2"
googledrive "no"       "4.5.2"
googlesheets4 "no"      "4.5.2"
gstat       "no"       "4.5.2"
[ reached 'max' / getOption("max.print") -- omitted 138 rows ]
> R.version
platform      x86_64-w64-mingw32
arch           x86_64
os             mingw32
crt            ucrt
system        x86_64, mingw32
status
major          4
minor          5.1
year           2025
month          06
day            13
svn rev        88306
language       R
version.string  R version 4.5.1 (2025-06-13 ucrt)
nickname       Great Square Root
> |
```

R version and packages:

```
> R.version

platform      x86_64-w64-mingw32
arch           x86_64
os             mingw32
crt            ucrt
system        x86_64, mingw32
status
major          4
minor          5.1
year           2025
month          06
day            13
svn rev        88306
language       R
version.string  R version 4.5.1 (2025-06-13 ucrt)
nickname       Great Square Root
> |
```

```
> installed.packages()
```

	Package	LibPath	Version	Priority	Depends
askpass	"askpass"	"D:/R/R-4.5.1/library"	"1.2.1"	NA	NA
backports	"backports"	"D:/R/R-4.5.1/library"	"1.5.0"	NA	"R (>= 3.0.0)"
base	"base"	"D:/R/R-4.5.1/library"	"4.5.1"	NA	NA
base64enc	"base64enc"	"D:/R/R-4.5.1/library"	"0.1-3"	NA	"R (>= 2.9.0)"
bit	"bit"	"D:/R/R-4.5.1/library"	"4.6.0"	NA	"R (>= 3.4.0)"
bit64	"bit64"	"D:/R/R-4.5.1/library"	"4.6.0-1"	NA	"R (>= 3.4.0), bit (>= 4.0.0)"
blob	"blob"	"D:/R/R-4.5.1/library"	"1.2.4"	NA	NA
boot	"boot"	"D:/R/R-4.5.1/library"	"1.3-31"	"recommended"	"R (>= 3.0.0), graphics, stats"
brew	"brew"	"D:/R/R-4.5.1/library"	"1.0-10"	NA	NA
brio	"brio"	"D:/R/R-4.5.1/library"	"1.1.5"	NA	"R (>= 3.6)"
broom	"broom"	"D:/R/R-4.5.1/library"	"1.0.10"	NA	"R (>= 4.1)"
bslib	"bslib"	"D:/R/R-4.5.1/library"	"0.9.0"	NA	"R (>= 2.10)"
BWstest	"BWstest"	"D:/R/R-4.5.1/library"	"0.2.3"	NA	NA
cachem	"cachem"	"D:/R/R-4.5.1/library"	"1.1.0"	NA	NA
callr	"callr"	"D:/R/R-4.5.1/library"	"3.7.6"	NA	"R (>= 3.4)"
cellranger	"cellranger"	"D:/R/R-4.5.1/library"	"1.1.0"	NA	"R (>= 3.0.0)"
class	"class"	"D:/R/R-4.5.1/library"	"7.3-23"	"recommended"	"R (>= 3.0.0), stats, utils"
cli	"cli"	"D:/R/R-4.5.1/library"	"3.6.5"	NA	"R (>= 3.4)"
clipr	"clipr"	"D:/R/R-4.5.1/library"	"0.8.0"	NA	NA
cluster	"cluster"	"D:/R/R-4.5.1/library"	"2.1.8.1"	"recommended"	"R (>= 3.5.0)"
codetools	"codetools"	"D:/R/R-4.5.1/library"	"0.2-20"	"recommended"	"R (>= 2.1)"
colorspace	"colorspace"	"D:/R/R-4.5.1/library"	"2.1-2"	NA	"R (>= 3.0.0), methods"
commonmark	"commonmark"	"D:/R/R-4.5.1/library"	"2.0.0"	NA	NA
compiler	"compiler"	"D:/R/R-4.5.1/library"	"4.5.1"	"base"	NA
conflicted	"conflicted"	"D:/R/R-4.5.1/library"	"1.2.0"	NA	"R (>= 3.2)"
cpp11	"cpp11"	"D:/R/R-4.5.1/library"	"0.5.2"	NA	"R (>= 4.0.0)"
crayon	"crayon"	"D:/R/R-4.5.1/library"	"1.5.3"	NA	NA
credentials	"credentials"	"D:/R/R-4.5.1/library"	"2.0.3"	NA	NA
curl	"curl"	"D:/R/R-4.5.1/library"	"7.0.0"	NA	"R (>= 3.0.0)"
data.table	"data.table"	"D:/R/R-4.5.1/library"	"1.17.8"	NA	"R (>= 3.3.0)"
datasets	"datasets"	"D:/R/R-4.5.1/library"	"4.5.1"	"base"	NA
DBI	"DBI"	"D:/R/R-4.5.1/library"	"1.2.3"	NA	"methods, R (>= 3.0.0)"
dbplyr	"dbplyr"	"D:/R/R-4.5.1/library"	"2.5.1"	NA	"R (>= 3.6)"
desc	"desc"	"D:/R/R-4.5.1/library"	"1.4.3"	NA	"R (>= 3.4)"
devtools	"devtools"	"D:/R/R-4.5.1/library"	"2.4.5"	NA	"R (>= 3.0.2), usethis (>= 2.1.6)"
diffobj	"diffobj"	"D:/R/R-4.5.1/library"	"0.3.6"	NA	"R (>= 3.1.0)"
digest	"digest"	"D:/R/R-4.5.1/library"	"0.6.37"	NA	"R (>= 3.3.0)"
downlit	"downlit"	"D:/R/R-4.5.1/library"	"0.4.4"	NA	"R (>= 4.0.0)"
dplyr	"dplyr"	"D:/R/R-4.5.1/library"	"1.1.4"	NA	"R (>= 3.5.0)"
dtplyr	"dtplyr"	"D:/R/R-4.5.1/library"	"1.3.2"	NA	"R (>= 4.0)"
ellipsis	"ellipsis"	"D:/R/R-4.5.1/library"	"0.3.2"	NA	"R (>= 3.2)"

运行结果:

```
> metadata <- read_csv("sample_metadata.csv")
Rows: 12 Columns: 5
— Column specification —————
Delimiter: ","
chr (4): sample_id, group, sex, batch
dbl (1): body_weight_g

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
> counts <- read_csv("toy_gene_counts.csv")
Rows: 8 Columns: 13
— Column specification —————
Delimiter: ","
chr (1): gene_id
dbl (12): S1, S2, S3, S4, S5, S6, S7, S8, S9, S10, S11, S12

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
> enzyme <- read_csv("enzyme_activity.csv")
Rows: 12 Columns: 4
— Column specification —————
Delimiter: ","
chr (2): sample_id, group
dbl (2): citrate_synthase_activity, lactate_mmol_L

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
>
> # Inspect the first rows
> head(metadata)
# A tibble: 6 × 5
  sample_id group sex batch body_weight_g
  <chr>      <chr> <chr> <chr>      <dbl>
1 S1        Control F B1         22.1
2 S2        Control M B1         24.5
3 S3        Control F B2         21.8
4 S4        Control M B2         25.2
5 S5        Control F B3         22.7
6 S6        Control M B3         24.9
> head(counts)
# A tibble: 6 × 13
  gene_id S1 S2 S3 S4 S5 S6 S7 S8 S9 S10 S11 S12
  <chr>   <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl> <dbl>
1 NA     101  140  129  124  133  119  244  229  243  218  223  232
```

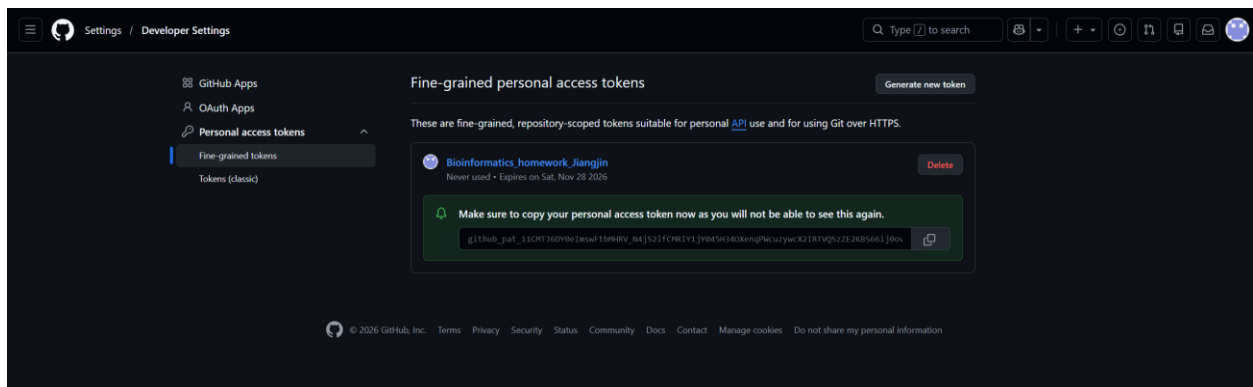
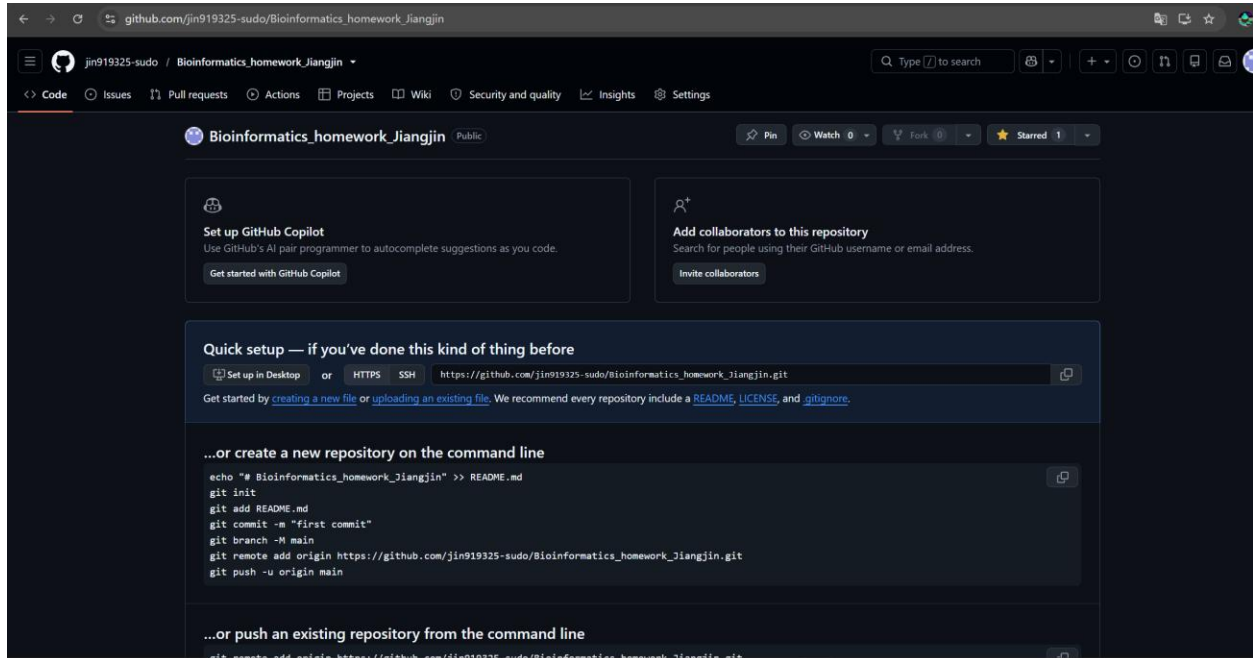
4. Install agent and create your first prompt log entry documenting AI use and verification.

4.1 Github:<https://www.github.com/> (Create your account and repository for your homework submission). You need an account and a repository for homework submission (following the pattern: Bioinformatics_homework_Yourname_ID). You also need personal access token in together with repository link to activate your EasyMultiProfiler-web account.

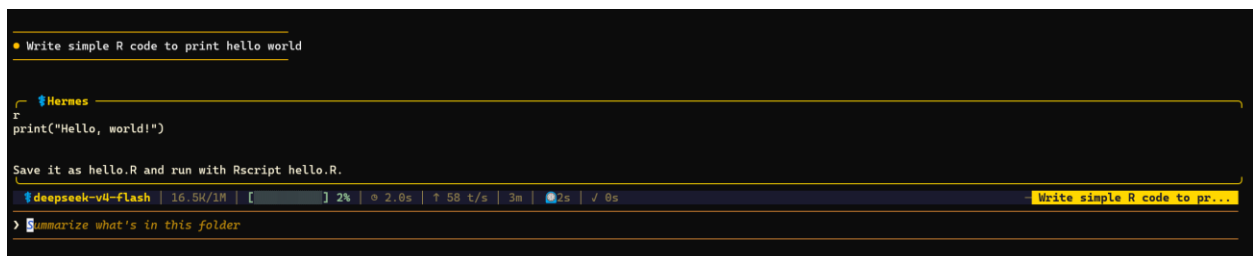
4.2 Hermes-agent: <https://github.com/NousResearch/hermes-agent> (Install agent for coding) and Hermes WebUI (<https://github.com/nesquena/hermes-webui>)

4.3 Codex: <https://chatgpt.com/codex/> (Install for coding and agent)

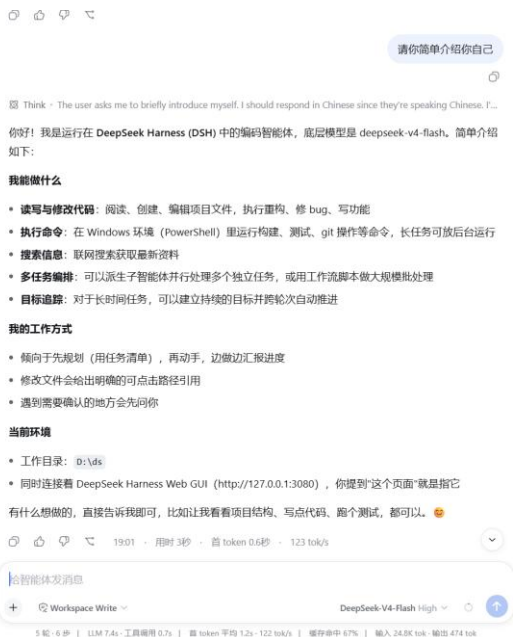
4.4 Deepseek API: <https://platform.deepseek.com/usage> (Top up ¥10 for agent set and switching to LLM accessed only on SUAT campus) or NVIDIA API



Hermes:



Deepseek:



5. Install EasyMultiProfiler and EasyMultiProfiler-Web for multi-omics data analysis and homework submission (You can use Hermes or Hermes WebUI to complete the installation step)

EasyMultiProfiler: <https://github.com/liubingdong/EasyMultiProfiler>

EasyMultiProfiler-Web: <https://github.com/xielab2017/EasyMultiProfiler-Web>

```
Windows PowerShell
git clone https://github.com/liubingdong/EasyMultiProfiler --> OK (exit 0)
git clone https://github.com/xielab2017/EasyMultiProfiler-Web --> OK (exit 0)

STEP 3 - Backend venv (C:\Users\jjj150\dev\EasyMultiProfiler\.venv)
py -3.12 -m venv .venv
./venv/Scripts/python.exe -m pip install --upgrade pip --> pip 26.2.1
pip list --> only pip. No third-party packages exist to install. EasyMultiProfiler is an R package (R >= 4.3.3), zero Python code in the repo. Its real dependencies are ~62 Imports + 19 Suggests R packages (McDNA, clusterProfiler, DESeq2, edgeR, vegan, xgboost, ...), which require R itself.

STEP 4 - Web venv (C:\Users\jjj150\dev\EasyMultiProfiler-Web\.venv)
py -3.12 -m venv .venv; pip upgraded to 26.2.1
pip install edge-tts pillow playwright --> edge-tts 7.2.8, pillow 12.3.0, playwright 1.62.0 (+ aiohttp deps)
Why only these: the web app's runtime (static_server.py, emp_gateway.py) is pure Python stdlib - zero pip deps. edge-tts/pillow/playwright are used only by 3 optional dev scripts (build_lesson_videos.py, render_anim_video.py, render_bio_video.py).
Verified: all repo .py files pass py_compile; static_server + emp_gateway import cleanly in the venv.

VERIFICATION SUMMARY
- Clones: 2/2 present and intact
- Backend venv: created, empty by design (no Python deps exist - documented above)
- Web venv: created, optional dev deps installed, runtime scripts verified importable

WHAT'S LEFT (the actual blocker): both repos' real dependencies are R packages, and R is NOT installed on this machine (no R, no Rscript, no Rtools). To actually run the platform you need:

1. Install R >= 4.3.3 (e.g. winget install RProject.R) and Rtools 4.4/4.5 (~480MB, needed to compile packages on Windows)
2. Backend: Rscript -e 'pak::pak("liubingdong/EasyMultiProfiler")' (or remotes::install_github)
3. Web: Rscript webapp/scripts/install_runtime.R (installs CRAN + Bioconductor deps from DESCRIPTION)

That's a large install (hundreds of R packages, likely 30-120+ min). Say the word and I'll run it.

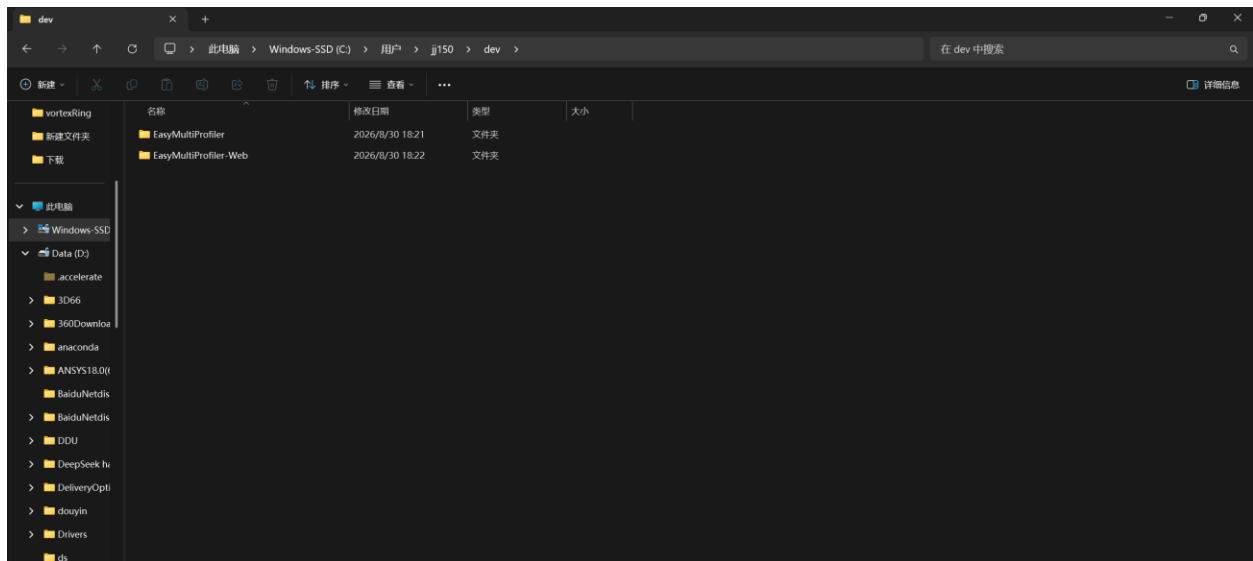
One note: playwright's browser binaries aren't downloaded (playwright install) - only needed if you actually run the video-generation scripts.

[ ] preparing memory...
memory +memory: "User's Windows 11 machine (home C:\Users\jjj150): 'python' on PATH is Hermes's own venv (no pip) - use 'py -3.12'/'py -3.13' launcher for real Python; Anaconda also at D:\anaconda. Node v24.20.0 + npm 11.19.0 installed. R and Rtools NOT installed (matters for R-based projects). Terminal is git-bash; native tools need C:/ paths."
0.0s

[ ] Hermes
Done. Repos are at C:\Users\jjj150\dev\EasyMultiProfiler, EasyMultiProfiler-Web, both with working Python 3.12 venvs. The one honest caveat stands: neither project's real dependency set is Python - it's R, and R isn't installed yet. Tell me if you want the full R setup run.

[ ] deepseek-v4-flash 53.6K/1M [ ] 5% [ ] 94.1% [ ] 10.4s [ ] 116 t/s [ ] 10m [ ] 9m 32s [ ] 0s Clone EasyMultiProfiler r...

[ ] lan a feature, then build it step by step
```



Homework submission

xielab2017

Bioinformatics_homework_XieLiwei

Code

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Pull requests

Agents

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Projects

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xielab2017

[EMP] sync Week_12/multomics/weekly (ChIPseq 联合分析) v8.0.0 run 202...

c7bc13 · last month

18 Commits

EMP2026

[EMP] sync Week_12/multomics/weekly (ChIPseq 联合分析)

last month

.gitattributes

Submit Homework_1: literature reference document

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.gitignore

Submit Homework_1: literature reference document

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README.md

Update course details in README.md

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README

Bioinformatics_homework_XieLiwei

Shenzhen University of Advanced Technology, Faculty of Synthetic Biology Course Name: Bioinformatics: From Multi-omics to Discovery Github Homework submission system

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1. Create your github account

2. Create your repository for this course. The naming style: Bioinformatics_homework_Yourname_ID

3. Scan QR code below and submit your infor